21 **D** Since the sphere is a conductor, the charge will reside only on the outermost surface of the hollow sphere and there is no charge inside the sphere. The electric field inside the sphere is thus zero, and the electric potential inside will be a constant,

$$V = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r},$$

which is equal to the potential at the surface of the sphere.

Outside the sphere, the potential V will decrease inversely with distance x measured from the centre of the sphere (i.e., $V \propto \frac{1}{x}$).